

When estimates of WTP are not available, it can be approximated by other measures, most notably COI measures.

An individual's WTP to avoid an adverse health effect will include, at a minimum, the amount of money he or she would have to pay for medical expenses associated with the illness. Because medical expenditures are to a significant extent shared by society, via medical insurance, Medicare, etc., the medical expenditures actually incurred by the individual are likely to be less than the total medical cost to society. The total value to society of an individual's avoidance of an adverse health effect, then, might be thought of as having two components: (1) the COI to society, including the total value of the medical resources used (some portion of which will be paid by the individual), plus the value of the lost productivity, as well as (2) the WTP of the individual, as well as that of others, to avoid the pain and suffering resulting from the illness.

The COI approach attempts to estimate the total value of the medical resources used up as well as the value of the individual's time lost as a result of the illness. Because this method does not include the value of avoiding the pain and suffering resulting from the illness (a potentially large component), it is generally believed to underestimate the total economic value of avoiding the illness, perhaps substantially.

The contingent valuation method (and conjoint analysis) attempts to elicit from people what they would be willing to pay (WTP) to avoid the illness. Because of the distortion in the market for medical goods and services, whereby individuals generally do not pay the full value of the medical care, this method too is likely to understate the total economic value of avoiding the illness.

Although the COI and WTP are the two most common methods, other methods have been used in certain circumstances. The method the benefit analyst chooses to value a particular health endpoint will depend in part on what data are available. The unit values currently available for use in BenMAP-CE are data or estimates that have been collected or generated by researchers and can be readily obtained in publicly available databases or in the open literature. When reviewing the economic literature to determine the appropriate valuation functions to use, it is important to have an economist assist.

7.1.1.2 Valuing Reductions in Premature Mortality

The economics literature discussing the value of changes in fatality risks is extensive and provides a basis for monetizing benefits when the number of deaths avoided as a result of an air quality improvement can be calculated, but the literature on certain issues regarding the appropriate method for valuing reductions in premature mortality risk is still developing. Issues such as the appropriate discount rate and whether there are factors, such as age or the quality of life, that should be taken into consideration when estimating the value of avoided premature mortality are still under discussion. BenMAP-CE currently offers a variety of options reflecting the uncertainty surrounding

the unit value for premature mortality. See the Appendices H and I for more detail on the valuation functions available in BenMAP-CE.

Monetary estimates of changes in premature mortality risk are often expressed in terms of the Value of a Statistical Life (VSL). This term is easily misinterpreted and should be carefully described when used in benefit analysis. VSL is the aggregate dollar amount that a large group of people would be willing to pay for a reduction in their individual risks of dying in a year, such that we would expect one fewer death among the group during that year on average. The basic assumption underlying the VSL approach is that equal increments in fatality risks are valued equally. For similar reasons, the VSL approach is only appropriate for marginal changes in the risk of death and should not be used to value more significant changes. Because changes in individual fatality risks resulting from environmental regulation are typically very small, the VSL approach is usually acceptable for these types of benefit analyses.

The U.S. EPA National Center for Environmental Economics provides answers to frequently asked questions regarding the economic value of mortality risk on its website: <https://www.epa.gov/environmental-economics/mortality-risk-valuation>. You may wish to consult this site as you have questions regarding how U.S. EPA derives VSL and applies it in an environmental benefits analysis.

7.1.2 Overview of Discounting

What is discounting?

In general, people prefer current consumption to future consumption. In other words, a \$1 today is worth more today than a \$1 tomorrow is worth today, and that dollar continues to decrease in value as you go further out into the future. (This concept is also referred to as the social rate of time preference or the time value of money. This is a different concept than inflation, which is a general increase in the price level of goods and services.) Discounting is the process of converting that future dollar into a value that can be compared to the value of a dollar today. The discount rate expresses this process in quantitative terms. The higher the discount rate, the faster value decreases over time. For example, \$1 twenty years from now is worth \$0.55 today at a 3% annual discount rate, but worth only \$0.26 at a 7% annual discount rate.

Fundamental Concept – Discounting / Discount Rate

In a cost-benefit analysis, **discounting** accounts for the fact that people value benefits that occur in the future less than benefits received today. The rate at which individuals discount the value of those benefits is the **discount rate**. Typically, if a benefit is expected to be realized as a stream of benefits over multiple years, as is often assumed for mortality risk reductions, the economic value of that benefit stream would be discounted back to the starting year of analysis and summed as a **Net Present Value** of benefits.

A basic discounting function is as follows:

$$\text{Present Value} = \text{Future Value} / (1+r)^t$$

where r is the discount rate and t is the time period (usually years).

Example: \$1 twenty years from now at a 3% annual discount rate is worth \$0.55 today

$$\text{Present Value} = \$1.00 / (1 + 0.03)^{20} = 1 / (1.03)^{20} = 1 / 1.806111 = 0.553676 = \$0.55$$

Why do we discount benefits?

The benefits of reductions in air pollution may need to be discounted for three key reasons:

1. Today's society values benefits that occur today more highly than benefits that will occur in the future. Therefore, we must discount in order to compare those future benefits with current benefits.
2. For a cost-benefit analysis, benefits estimates in a future year need to be comparable to the cost estimates for that same year (which are also discounted).
3. Discounting can be used to compare the future *streams* of benefits and costs. The core BenMAP-CE program estimates changes in adverse health effects based on changes in air quality for one specified analysis year, even though certain health benefits may occur after the analysis year.²⁴ Discounting can be used to compare the future benefits with benefits occurring during the analysis year.

Under which scenarios would I need to discount benefits?

Health benefits may occur three different ways after the analysis year specified in BenMAP-CE.

1. Certain health endpoints accrue medical expenses or lost earnings for multiple years. The future medical expenses would need to be discounted to compare with expenses occurring in the analysis year.
2. Pollution exposure and the resulting health effects do not occur within the same year (a.k.a. a cession lag). The monetized benefits of future health effects would need to be discounted to compare with the benefits of health effects that occur during the analysis year.
3. In some analyses, you may want to estimate a stream of benefits occurring over multiple analysis years instead of just one analysis year. In this scenario, you would need to discount the future benefits occurring in each year analyzed back to the present year in order to present the cumulative total estimate of benefits (i.e., the net present value of a stream of benefits).

²⁴ The PopSim tool estimates the change in population mortality risk over a multi-year period, but it is not yet possible to estimate the economic value of these impacts in BenMAP-CE.

When would we not discount benefits?

In many instances, it is not necessary to discount the benefits estimates generated by BenMAP-CE. If the health effect and the monetized value of all the medical expenses, lost earnings, and suffering occur entirely in the analysis year, then you may not need to discount your benefits. For example, school loss days occur within the analysis year, and all monetized expenses occur within the analysis year. It is important that you understand the assumptions within the health and valuation functions before you decide whether you need to discount. (If your analysis year for your benefits estimates does not match the analysis year for your costs estimates, you may need to discount in order to compare your benefits with your costs even if you meet the criteria listed above.)

Which discount rate should I choose?

Selecting a discount rate is challenging and is one of the most contentious methodological issues encountered in economic analyses of environmental policies. Because environmental regulations frequently have differing streams of costs and benefits over time, the selected discount rate may determine whether the benefits of a regulatory action exceed the costs. In addition, selecting a higher discount rate may result in a smaller benefits estimate because the future benefits are worth much less than they would be if a lower discount rate was selected.

There are various economic arguments in support of and in opposition to various discount rates. To comply with OMB and EPA's recommendations, EPA currently uses discount rates of 3% and 7% for benefit analyses. For more details, see EPA [1999; 2000] listed in Chapter 1, Section 1.7 (Sources for More Information).

Which health endpoints accrue medical expenses or lost earnings for multiple years, and how do I discount them?

BenMAP-CE includes health and valuation functions for several chronic health effects, including PM_{2.5}-related chronic bronchitis and non-fatal acute myocardial infarctions (AMIs, or heart attacks).

- Chronic bronchitis is assumed to last from the initial onset of the illness throughout the rest of the individual's life. BenMAP-CE currently includes one WTP function as well as two COI functions representing the two discount rates for chronic bronchitis.
- Technically, AMIs are discrete, acute events, not chronic conditions. However, heart attacks cause chronic follow-up health effects that accrue medical expenses over time, similar to chronic conditions. You can discount the economic value of these chronic effects through the valuation function in BenMAP-CE. Although WTP functions for AMIs are not available, BenMAP-CE currently includes several COI functions that incorporate the direct medical costs and the opportunity cost (lost earnings) for specific age groups at two discount rates.